

**Student ID:** \_\_\_\_\_

CSE 329 Machine Learning

CT 1

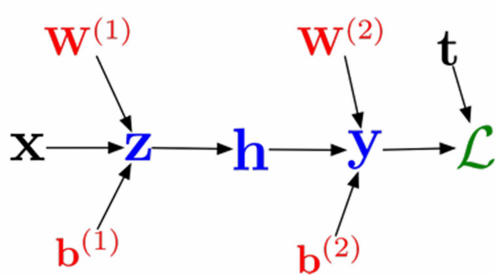
Time: 25 Minutes

Marks: 20

1. Derive the parameter update rule in logistic regression that maximizes the likelihood of the parameters. Show the derivation step by step. [6]

2. State the normal equation corresponding to the optimal parameter estimate in linear regression (The derivation is **not** requested). Discuss two limitations of the closed-form solution of linear regression. [2 + 2]

3. Given the computation graph and the functions below, derive the  $\overline{W^{(1)}} = \frac{d\mathcal{L}}{dW^{(1)}}$ . Assume all notations carry their usual meanings. [6]



$$\mathbf{z} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$$

$$\mathbf{h} = \sigma(\mathbf{z})$$

$$\mathbf{y} = \mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)}$$

$$\mathcal{L} = \frac{1}{2}\|\mathbf{t} - \mathbf{y}\|^2$$

4. Provide two examples of common activation functions with their mathematical expressions and the corresponding function diagrams. [2 + 2]